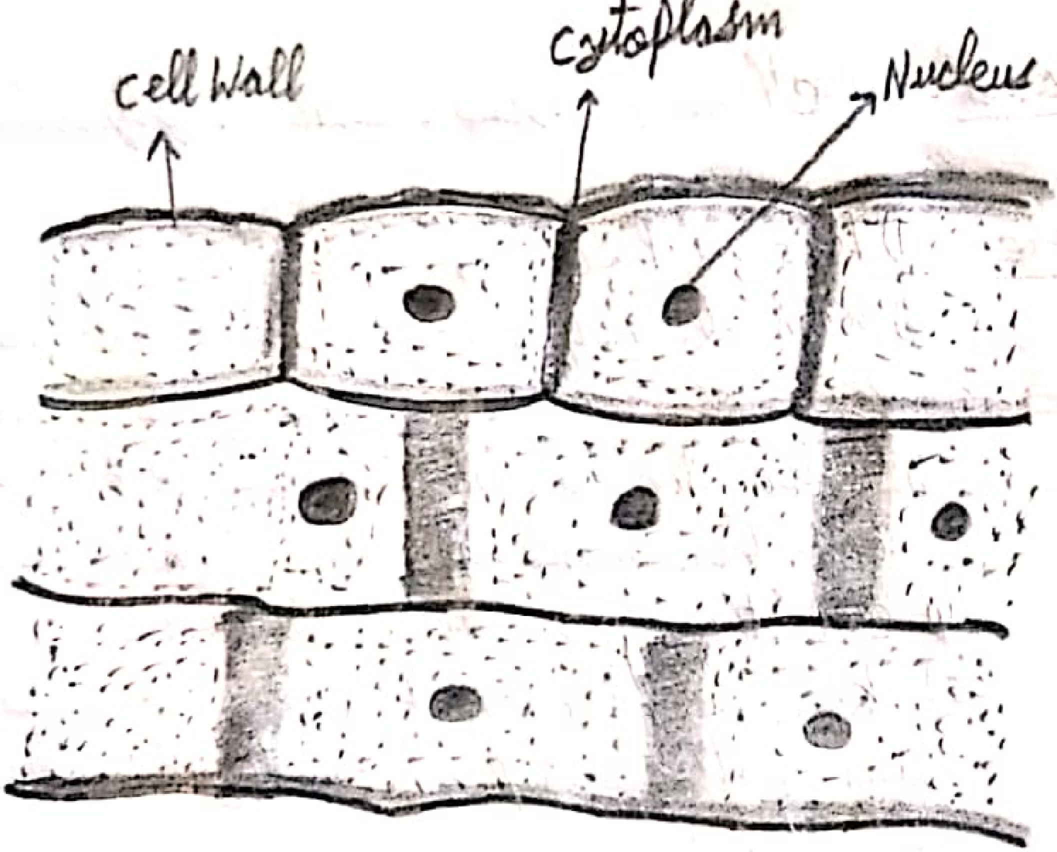
# DIAGRAM



Experiment No. 2(a) Page No. Date 06/08/26

→ To Prepare and study Plant cells from Onion Peel.

→ **Requirement**- Microscope, cover slips, Watch Glass, slide, Blade, onion, forceps, Muslin cloth, blotting paper

→ **Chemicals**- Methylene blue, water, glycerine.

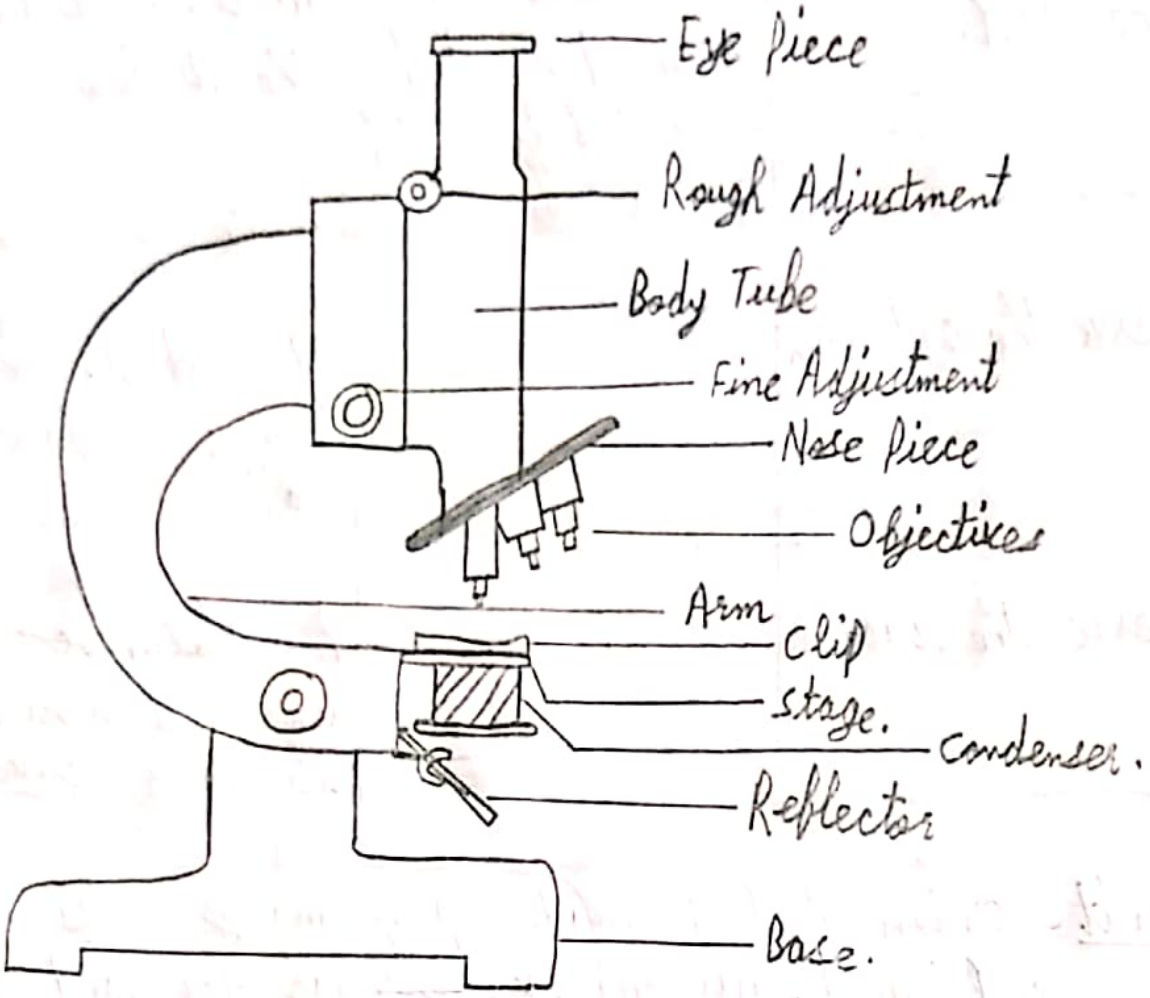
→ **Procedure**- 1) Take a Piece of Onion.
2) With the help of forceps remove the white layer or Onion Peel from its concave side.
3) Place the removed onion Peel into water and then make it dry with muslin cloth
4) Cut this Onion Peel into Pieces.
5) keep a piece of peel on the glass slide and put a drop of methylene blue stain on it.
6) Put a drop of glycerine on stained Peel and place a cover slip on it
7) Clean the slide from edges.
8) Observe this mount first under low Power and then under hight Power microscope.

| Observation | Result | |
|---|---|---|
| | Low Power Microscope | High Power Microscope |
| Observe the shape of the cell | - Shape of the cell will be rectangle | - Shape of the cell will be rectangular with distin-ct cell wall. |

| | | | |
|---|---|---|---|
| - | Observe the nucleus of the cell. | There is a deep colored central round body, which is nucleus. | Nucleolus is visible within the Nucleus. |
| - | Observe the cytoplasm | ------------ | It is deep coloured on inner surface of cell Wall. |
| - | Observe the vacules | -------------- | Protoplasm contains three vacuoles which has its outer covering. |

- **Result**- Onion Peel is made up of many cells having cell wall, nucleus, ~~vacuole~~ vacuoles and cytoplasm.

# DIAGRAM




Compound Microscope.

Teacher's Signature........................

# AFZAN LAB MANUAL

Experiment No. 01 Page No. Date 15-08/26

⇒ <u>To study the main parts of microscope:</u>

• <u>Requirement:</u>- A compound microscope.

⇒ <u>Main Parts of Compound Microscope:</u>

1) <u>Base:</u> It is the metallic, very heavy and horse-shoe shaped structure which bears the whole weight of the microscope and provides stability to the microscope

2) <u>Pillar:</u> It is the strong, vertical structure which gives support to the arm.

3) <u>Arm:</u> It is curved part which is used to hold the microscope.

4) <u>Stage:</u> A strong, metallic plate is fixed at the lower end of the ~~stage~~ arm, known as stage.

5) <u>Stage Clips:</u> Two clips are attached to the stage to hold the slide.

6) <u>Diaphragm:</u> It is fitted below the stage to regulate the amount of light entering the microscope.

7) <u>Condenser:</u> It is also present below the stage to concentrate the light rays.

8) <u>Body Tube:</u> It is a circular, hollow structure attached to the upper end of the arm. It can be moved up and down.

9) <u>Nose Piece:</u> It is a ~~circle circle~~ circular, revolving disc attached to lower end of body tube and usually, bears three objective lenses of different magnification

10) <u>Draw Tube:-</u> It is a tube which can move in and out of the body tube and bears an eye piece at the top.

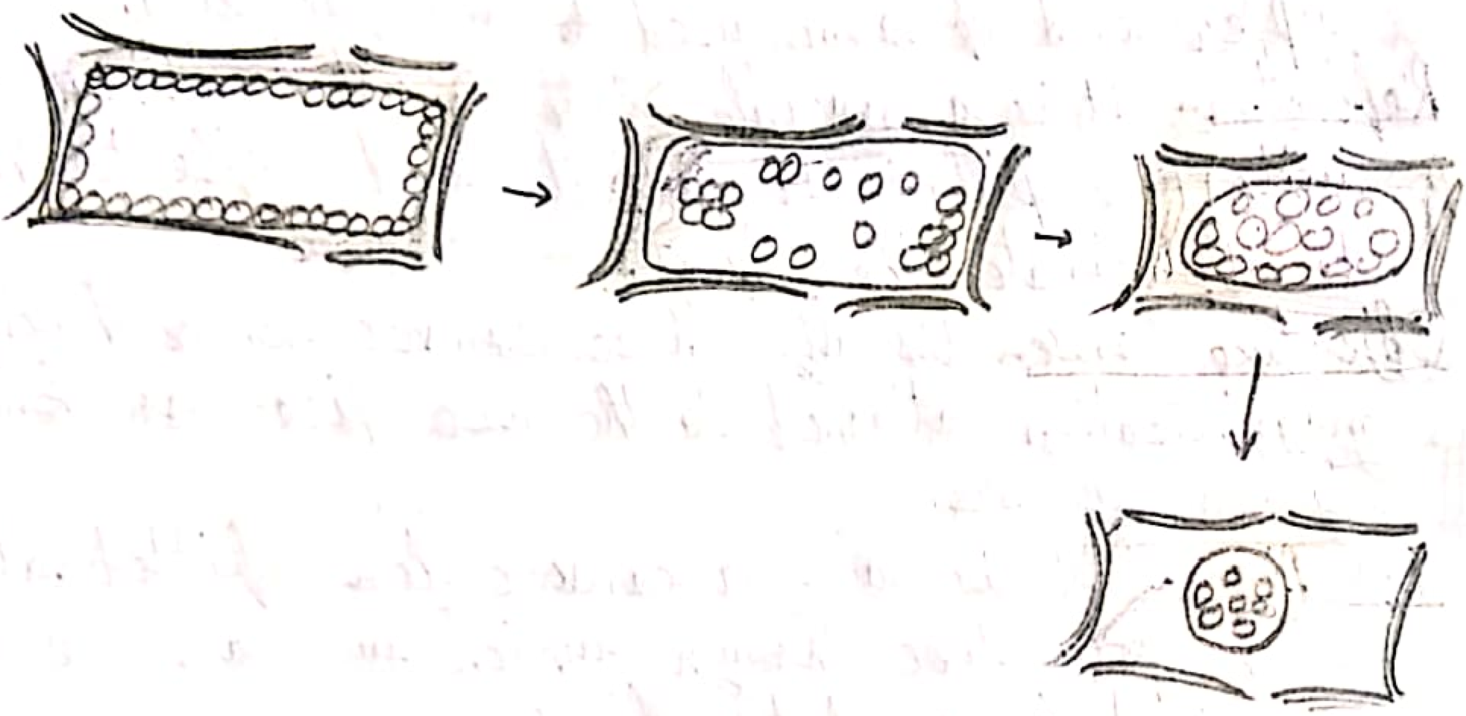
11) <u>Coarse Adjustment Screw.</u> It is a bigger sized screw at the upper end of arm and is used to move the body tube up and down for focusing the object.

12) <u>Fine Adjustment Screw:</u> It is a smaller sized screw at the upper end of arm used ~~to~~ fine focussing.

13) <u>Reflector.</u> It is a movable double sided mirror attached with the pillar. It collects and reflects the light to illuminate the object.

14) <u>Objective lense.</u> Usually three convex lenses having diffrent magnification attached to the nose piece are known as objective lenses.

15) <u>Eye Piece.</u> It is also a convex lens fitted at the top of the body tube through which we can see the magnified image of the object.



# AFZAN LAB MANUAL

Experiment No. 3(b) Page No. Date 15-08-26

⇒ To Demonstrate the Plasmolysis and deplasmolysis in Plant cells:

⇒ Requirement: Onion bulb, petridish, slide, coverslips, forceps, needle, brush, microscope and sucrose.

⇒ Procedure:
1) Take an onion bulb and cut it into four pieces.
2) Take one piece, remove one scale thick from it and tear ~~of~~ it from concave side.
3) With the help of forceps, remove the peel from it.
4) Put this peel in water and then cut it into small pieces.
5) Put this piece of peel on dry, clean slide and spread it carefully with the help of needle and brush insuch a way that there should be no wrinkle on it and cover it with coverslip.
6) Observe it carefully under the microscope and make its ~~sketh stetch~~ sketch.
7) Add few drops of concentrated sucrose solution through the sides of the slide so that it reaches to peel.
8) Examin the peel again under microscope.
9) Now drain out the concentrated sucrose solution, add few drops of water to it and again examine it under microscope carefully.

⇒ Observation: When concentrated Sucrose (sugar) solution is added to the slide, then protoplasm of cells shrink away from the cell wall, leaving the cell wall in its position

→ Conclusion:- Concentration of water is higher inside the cell than outside due to the absorption of water by sucrose solution outside the cell and hence due to osmosis cell loses water i.e plasmolysis occur.

# AFZAN LAB MANUAL

Experiment No. 4 Page No. Date 24/08/26

→ To study different types of tissue from permanent slides/model of Parenchyma, Collenchyma, Sclerenchyma.

→ **Requirement** - A compound microscope, permanent slides of Parenchyma, Collenchyma, Sclerenchyma.

→ **Procedure** - Study the permanent slides of Parenchyma, Collenchyma, Sclerenchyma one by one under the microscope.

→ (A) **Parenchyma** -
1) It consists of isodiametric living cells which may be oval, polygonal or rounded in shape.
2) They have thin cell wall made up of cellulose.
3) Each cell has a large central vacuole.
4) They have small inter cellular spaces.
5) It is the most abundant tissue in plants.
6) They occur in soft parts of stems, leaves, roots, flowers etc.
7) Their cytoplasm contains a small nucleus.

→ (B) **Collenchyma** -
1) Collenchyma cells are living cells which are usually elongated with thick corners.
2) They have no inter cellular spaces.
3) Each cell possesses a large central vacuole.
4) Collenchyma cells may be oval, circular or polygonal in shape.
5) These are found in the ~~tw~~ leaves, leaf, stalks and stems.

6) Usually few chloroplasts are present in the ~~of~~ collenchyma cells.

(C) Sclerenchyma- 1) These cells are highly elongated, narrow and spindle shaped.
2) They have no inter cellular spaces.
3) They have lignified cell walls.
4) They have dead cells and hence no protoplasmic content.
5) These are found in stems, roots, hard coverings of seeds and nuts.